

Running the database: the fuller list

Governance

- **A named data steward**, with a written mandate and a deputy, and a **network of validators** per species group. In Czechia the Nature Conservation Information System is set up in law (Act No. 364/2021 Coll.), which names its operator and its purposes, EU reporting among them. Here: *legal basis and steward – to be confirmed*.
- **Data-sharing agreements** with every institution that sends or receives data: which data, how often, in what format, under which licence, and how sensitive records are handled.
- **A sensitive-species policy** with an owner and a review cycle: generalise the location in public; full precision for named users under agreement.

Procedures

- **SOPs and a data dictionary**. One written procedure for each flow (monitoring, trusted experts, citizen science); one definition for each field, with its Darwin Core term – the crosswalk opposite is its core.
- **A data management plan for every funded project**: format, schedule, licence and where the data are deposited – agreed at contract signature, checked at hand-over. The data, not the report, are the deliverable.
- **Raw data never overwritten**; provenance and a change history on every record. Persistent **occurrenceID** and **eventID**, never reused.
- **Backups** that run automatically, with at least one copy off-site, and a restore that has actually been tested.

Publishing and people

- **Metadata for every dataset in EML** (Ecological Metadata Language) – who, what, where, when, how, licence. GBIF’s IPT writes it for you.
- **Clear licensing and citation** – CC0, CC BY or CC BY-NC, stated per dataset; a GBIF DOI gives each dataset a citable identifier.
- **Regular public exports** – a versioned snapshot at fixed intervals, so an analysis can be repeated on the same data.
- **Training for data providers** – the forms, the checklist, what coordinate uncertainty means – and feedback to them on their own records.
- **Start simple, then iterate**: a checklist, one validated flow, one dataset on GBIF; then the next flow.

Which standard, and how settled is it?

Piece	Status
Darwin Core, Occurrence and Event cores	TDWG standard since 2009, maintained
Measurement or Fact (<i>dwc:</i>)	part of Darwin Core
Extended Measurement or Fact	from OBIS; registered with GBIF, not a TDWG standard
Humboldt Extension (<i>eco:</i>)	TDWG-ratified 28 Feb 2024; in GBIF’s registry for the Event core; tool support still growing
GBIF taxonomy	now the Catalogue of Life eXtended Release; the old GBIF Backbone is frozen (last built 2023)

Key resources

- Darwin Core terms – <https://dwc.tdwg.org/terms/>
- Humboldt Extension – <https://eco.tdwg.org/>
- Extended Measurement or Fact – https://manual.obis.org/format_emof.html
- GBIF IPT – <https://www.gbif.org/ipt> · manual <https://ipt.gbif.org/manual/>
- GBIF metadata profile (EML) – <https://ipt.gbif.org/manual/en/ipt/latest/gbif-metadata-profile>
- Catalogue of Life – <https://www.catalogueoflife.org/>
- PESI, the European taxonomic backbone – <https://www.eu-nomen.eu/portal/>
- Art. 17 reporting (Eionet) – https://cdr.eionet.europa.eu/help/habitats_art_17
- *ndopred*, Red List metrics from occurrence records – <https://github.com/jonasgaigr/ndopred>
- This workshop – <https://github.com/jonasgaigr/kosovo-biodiversity-data-workshop-2026>

Kosovo* – *This designation is without prejudice to positions on status, and is in line with UNSCR 1244/1999 and the ICJ Opinion on the Kosovo declaration of independence.

Full crosswalk: survey spreadsheet → Darwin Core

One visit is an **event**; what was found on it are **occurrences**; anything measured hangs from either as a **measurement**. Terms without a prefix are Darwin Core; **eco:** is the Humboldt Extension; **dcterms:** is Dublin Core, which Darwin Core uses for licence and rights.

Your column	Darwin Core term	Note
Event – the visit		
Visit no.	eventID	persistent, never reused
Site / plot code	locationID	from the site register
Date, time	eventDate	ISO 8601: 2026-06-18T21:30
Latitude, longitude	decimalLatitude decimalLongitude	decimal degrees
Coordinate system	geodeticDatum	e.g. EPSG:4326 (WGS 84)
GPS accuracy (m)	coordinateUncertaintyInMeters	mandatory
Place name	locality	as written in the field
Method	samplingProtocol	from a controlled list
Plot size	sampleSizeValue sampleSizeUnit	e.g. 25 · square metre
Time spent, traps set	samplingEffort or eco:samplingEffortValue eco:samplingEffortUnit	Humboldt splits value and unit
Target group	eco:targetTaxonomicScope	what was searched for
All targets recorded?	eco:isAbsenceReported eco:isTaxonomicScopeFullyReported	what makes an absence usable
Surveyor(s)	recordedBy eco:samplingPerformedBy	person; team or body
Occurrence – what was found		
Record no.	occurrenceID	persistent, never reused
Species	scientificName taxonID	national checklist ID; crosswalked to PESI and GBIF
Found? yes / no	occurrenceStatus	present / absent
Count	individualCount or organismQuantity organismQuantityType	e.g. 12 · individuals
Life stage, sex	lifeStage sex	controlled values
Identified by	identifiedBy	
Checked by an expert?	identificationVerificationStatus	free text in DwC – agree a national list
Record type	basisOfRecord	HumanObservation, PreservedSpecimen ...
Project, source	datasetName datasetID	keeps the source visible
Location generalised?	dataGeneralizations informationWithheld	for sensitive species
Remarks	occurrenceRemarks	
Measurement – linked by eventID or occurrenceID		
Water depth, cover %, ...	measurementType measurementValue measurementUnit	Extended Measurement or Fact
Dataset – once, in the EML metadata		
Licence	dcterms:license	e.g. CC BY 4.0
Rights holder	dcterms:rightsHolder	

To fill in locally

National checklist owner: _____

Data steward: _____

Validators per species group: _____